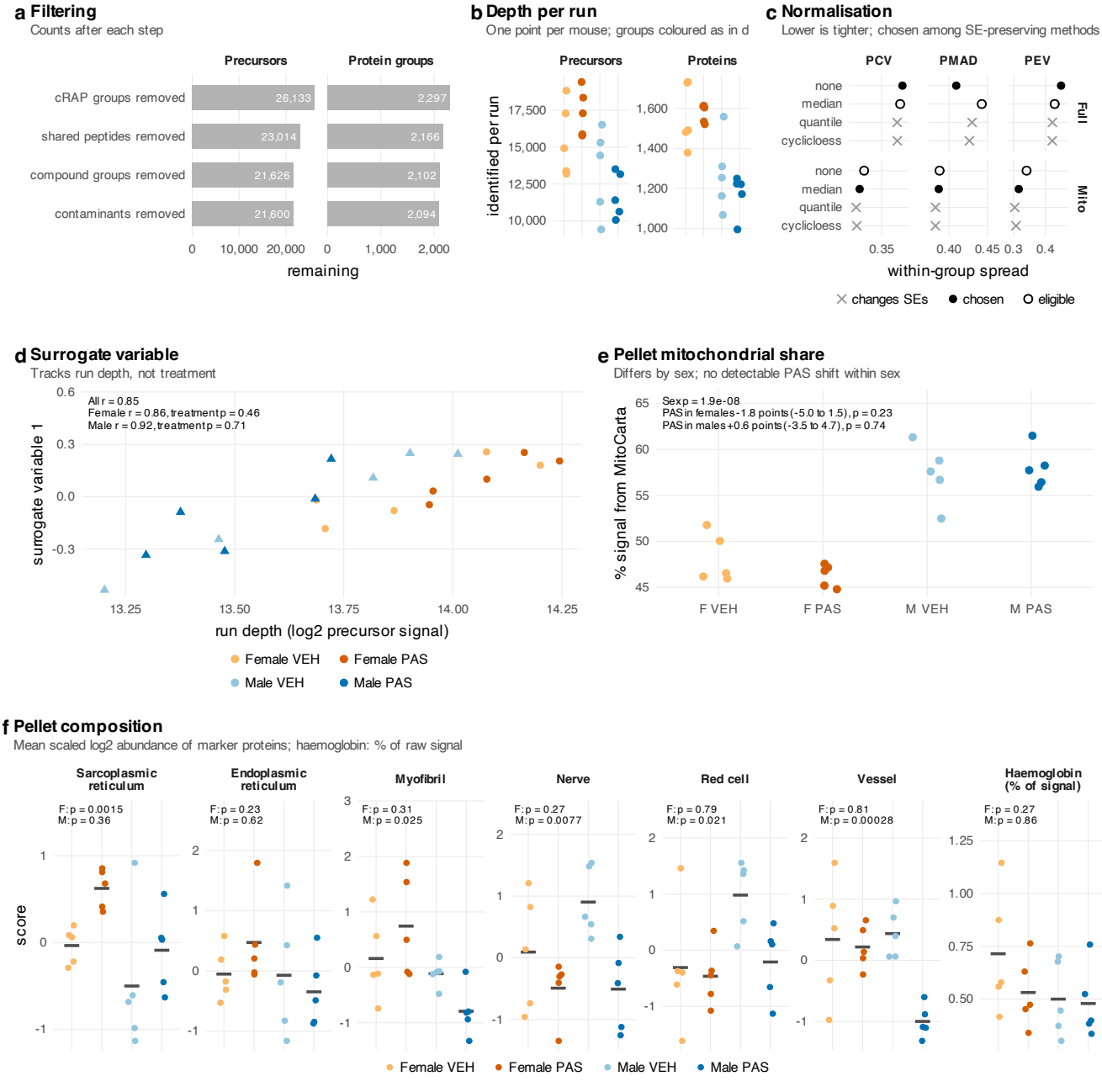
a Filtering
Counts after each step
Precursors
Protein groups
cRAP groups removed
shared peptides removed
compound groups removed
contaminants removed
26,133
23,014
21,626
21,600
2,297
2,166
2,102
2,094
0
10,000
20,000
1,000
2,000
remaining
b Depth per run
One point per mouse; groups coloured as in d
Precursors
Proteins
identified per run
17,500
15,000
12,500
10,000
1,600
1,400
1,200
1,000
c Normalisation
Lower is tighter; chosen among SE-preserving methods
PCV
PMAD
PEV
Full
Mito
none
median
quantile
cyclicloess
0.35
0.40
0.45
0.3
0.4
within-group spread
changes SEs
chosen
eligible
d Surrogate variable
Tracks run depth, not treatment
All r = 0.85
Female r = 0.86, treatment p = 0.46
Male r = 0.92, treatment p = 0.71
surrogate variable 1
0.6
0.3
0.0
-0.3
13.25
13.50
13.75
14.00
14.25
run depth (log2 precursor signal)
Female VEH
Female PAS
Male VEH
Male PAS
e Pellet mitochondrial share
Differs by sex; no detectable PAS shift within sex
Sex p = 1.9e-08
PAS in females -1.8 points (-5.0 to 1.5), p = 0.23
PAS in males +0.6 points (-3.5 to 4.7), p = 0.74
% signal from MitoCarta
65
60
55
50
45
F VEH
F PAS
M VEH
M PAS
f Pellet composition
Mean scaled log2 abundance of marker proteins; haemoglobin: % of raw signal
Sarcoplasmic reticulum
F: p = 0.0015
M: p = 0.36
Endoplasmic reticulum
F: p = 0.23
M: p = 0.62
Myofibril
F: p = 0.31
M: p = 0.025
Nerve
F: p = 0.27
M: p = 0.0077
Red cell
F: p = 0.79
M: p = 0.021
Vessel
F: p = 0.81
M: p = 0.00028
Haemoglobin (% of signal)
F: p = 0.27
M: p = 0.86
score
Female VEH
Female PAS
Male VEH
Male PAS